Journal of Anxiety Disorders 100 (2023) 102791

Contents lists available at ScienceDirect

Journal of Anxiety Disorders

journal homepage: www.elsevier.com/locate/janxdis

The role of fear of evaluation in group perception

Jin Shin [a,*], Thomas L. Rodebaugh [b]

[a] Department of Psychological & Brain Sciences, Washington University in St. Louis, USA
[b] University of North Carolina at Chapel Hill Department of Psychology and Neuroscience, USA

ARTICLE INFO

Keywords:
Bivalent fear of evaluation
Fear of Negative Evaluation
Fear of positive evaluation
Social Anxiety
Warmth
Dominance

ABSTRACT

Social anxiety disorder (SAD) is associated with interpersonal impairment. One possible reason for this dysfunction is that people with SAD evaluate others differently on dimensions of warmth and dominance compared to individuals without the disorder. In the current study, we examined whether two core constructs of SAD, fear of negative evaluation and fear of positive evaluation, affect the judgments that people make about groups based on warmth and dominance. We also investigated whether racial similarity (i.e., whether someone is the same race as those they're interacting with) and ethnic identity (i.e., one's sense of belonging to a particular social group) played a role in the types of evaluations people made. We created vignettes about groups varying in warmth and dominance, as well as photos varying in racial makeup. We presented photo-vignette pairs to participants and asked them to rate their desire to interact with the groups depicted in the photo-vignette. Participants in general reported greater desire to interact with warmer and less dominant groups. People with higher fear of negative evaluation reported higher desire for interaction with warmer groups, and those with higher fear of positive evaluation reported higher desire to interact with less dominant groups. We did not find any support for our hypothesis that people with stronger ethnic identity would show greater desire to interact with groups that were more similar to their race. Implications for treatment and directions for further research are discussed.

1. The role of fear of evaluation in individuals' perceptions of groups

Research suggests that both fear of negative evaluation (FNE) and fear of positive evaluation (FPE) are core constructs of social anxiety disorder (SAD) (Rodebaugh et al., 2012; Weeks & Howell, 2012). Another core feature of SAD is interpersonal impairment across multiple domains, including relationships with peers, friends, and romantic partners (Ruscio et al., 2008; Tonge et al., 2020; Weisman et al., 2011). One possible explanation for this impairment is that individuals with SAD evaluate others differently compared to individuals without the disorder. Indeed, research on impression formation suggests that individuals with SAD rate others differently on social rank and friendliness compared to people without SAD (Aderka et al., 2013).

1.1. Social anxiety and interpersonal evaluations

One widely-used model for understanding interpersonal behavior and reaction formation is the interpersonal circumplex (T. Leary, 1957; Wiggins, 1979). The interpersonal circumplex is made of two orthogonal axes, with affiliation or warmth represented on the horizontal axis, and power or dominance on the vertical axis. Using this model, a given interpersonal behavior can be conceptualized as having some combination of warmth (versus coldness) and dominance (versus submissiveness). The interpersonal circumplex can be partitioned into different sections; for our study, we used a circumplex with eight octants, which has been used in previous research on social anxiety (Rodebaugh et al., 2016). As one progresses around the circumplex, each octant represents a combination of the dimensions of warmth and dominance (see Fig. 1). Previous studies focusing on individuals with higher social anxiety have used these dimensions of warmth and dominance to examine interpersonal behavior, and found that people with SAD evaluate others differently than people without SAD (e.g., Aderka et al., 2013; Haker et al., 2014).

However, most of the research on SAD and interpersonal evaluations has focused on how people with SAD evaluate individuals (Aderka et al.,

* Correspondence to: Department of Psychological and Brain Sciences, Washington University in St. Louis, One Brookings Drive, Campus Box 1125, St. Louis, MO 63105, USA.

E-mail address: jin.shin@wustl.edu (J. Shin).

https://doi.org/10.1016/j.janxdis.2023.102791
Received 26 April 2023; Received in revised form 8 September 2023; Accepted 30 October 2023
Available online 1 November 2023

2013; Haker et al., 2014; Rodebaugh et al., 2016). Yet, many people with SAD struggle in both one-on-one and group interactions (American Psychiatric Association, 2022), indicating that data regarding how individuals with SAD evaluate groups would also be useful. Furthermore, there is little literature examining how core constructs of SAD, such as FNE and FPE, relate to evaluating others. Given that the psychoevolutionary theory underlying FNE and FPE is linked to group membership (i.e., FNE is tied to fear of being ostracized from one's group if evaluated negatively and FPE is related to fear of moving too quickly up the group's hierarchy and being in a position of increased competition with group members that feel threatened) (Gilbert, 2001), investigation into the relationship between fear of evaluation and interpersonal evaluations of groups seems particularly relevant.

Although one might assume that FNE and FPE would exhibit similar relationships as SAD to evaluations of others, this may not necessarily be the case. Research shows that FNE and FPE are highly correlated constructs but that each is uniquely and separately related to SAD (Rodebaugh et al., 2012; Wang et al., 2012; Weeks & Howell, 2012). Furthermore, researchers have found that the direction and strength of the relationship between FNE and other SAD-related constructs, such as positive and negative affect, is different from that of the direction and strength of the relationship between FPE and those same social anxiety-relevant constructs (Weeks et al., 2008, 2010). Taken together, the literature suggests that FNE and FPE are independent and separable from one another and thus may play different roles in impression formation relevant to SAD.

1.2. Interpersonal evaluations and perceived similarity

In line with the previously mentioned studies, research by Rodebaugh et al. (2016) suggests that people with high social anxiety show more desire to interact with less warm and less dominant individuals compared to people with low social anxiety. Further analyses showed that perceived similarity explained the bulk of these effects in that people demonstrated a higher desire to interact with others who were more similar to themselves. In other words, compared to those with low social anxiety, people with high social anxiety perceived themselves as less warm and less dominant, and showed more desire to interact with others who were less warm and less dominant compared to people with lower social anxiety.

The study by Rodebaugh et al. (2016), which examined the effects of perceived similarity on desire for interaction among individuals with social anxiety, lays the foundation for some follow-up questions. For example, does perceived similarity only apply to similarity in warmth and dominance, or can it apply to other similarities, such as characteristics related to race or ethnicity? Are interpersonal evaluations of warmth and dominance different if the race of the person that the individual with SAD is interacting with is the same or different? These may be important questions to investigate, as it is plausible that dynamics between racial majority and racial minority group members may be contributing factors to perceptions of threat or dominance (Sidanius & Pratto, 1999). Furthermore, members of a given social category could evaluate someone more favorably depending on whether that individual is viewed as part of the ingroup (i.e., same social category) or less favorably if viewed as part of the outgroup (i.e., different social category) (Brewer, 2017).

In addition to examining the relationship of perceived race to interpersonal evaluations, it is worth considering whether the related construct of ethnic identity plays a role in the evaluation of others. Ethnic identity is an aspect of an individual's self-concept that is defined by their sense of membership to a social or cultural group (Phinney & Ong, 2007). Indeed, research conducted in New Zealand shows that for individuals of certain ethnic minority groups, ethnic identity, specifically how strongly they identified as part of their ethnic group, may play a role in evaluating ingroup members more positively than outgroup members (Hamley et al., 2020). However, to our knowledge, ethnic identity as it relates to interpersonal evaluations in the context of SAD has not yet been explored, even though it is highly plausible that

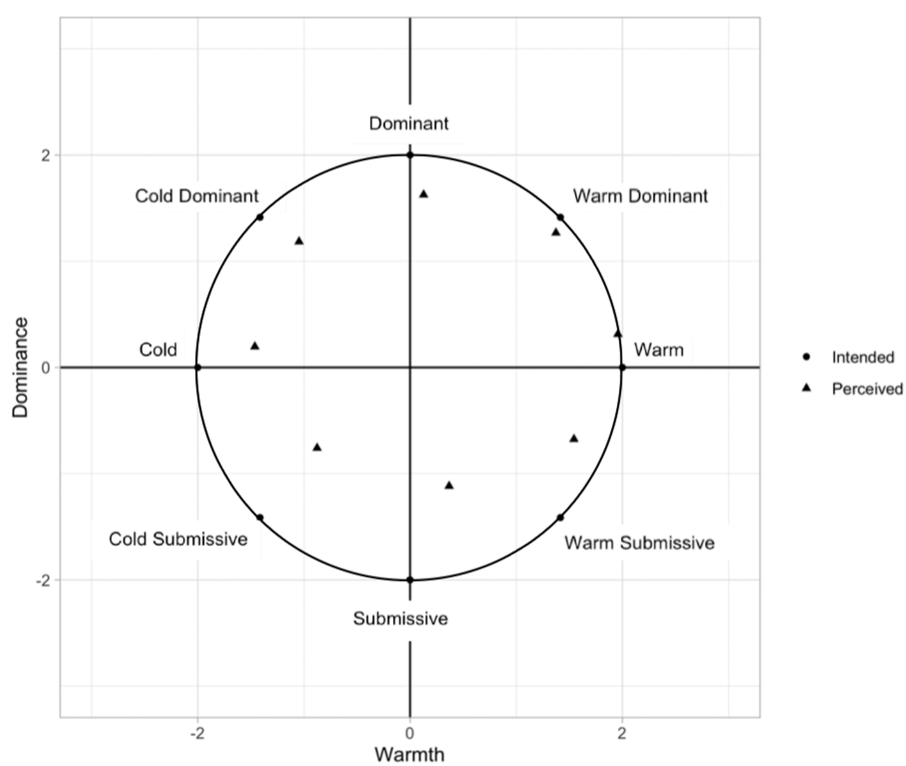

Fig. 1. *Manipulation Check Comparing Intended Vignette Ratings and Perceived Vignette Ratings.* Note. The intended points refer to the octants that the vignettes were intended to represent, based on the octant names in the interpersonal circumplex. The perceived points are estimated from the ratings that the pilot participants provided.

ethnic identity would affect one's perception of others. Furthermore, there is evidence that fears of evaluation operate differently across collectivistic and individualistic cultures (Okawa et al., 2021). These differences may be due to the prioritization of different values (e.g., modesty in collectivistic cultures makes FPE more adaptive than in individualistic cultures) (Okawa et al., 2021). Thus, it may be important to examine ethnic identity in the context of interpersonal evaluations, as cultural values may contribute to differences in how people react to others.

Currently, there is limited research examining SAD in the context of same-race and different-race interactions. In one study based in South Africa, researchers found that ethnicity influenced severity of symptoms related to social anxiety. Interestingly, Black participants reported higher SAD-related distress when interacting with other Black individuals than when interacting with people of a different race (Jager et al., 2014). To our knowledge, this is the only study examining SAD, interpersonal interactions, and racial similarity, highlighting how this topic has been largely ignored in the literature. The dearth of research regarding SAD-related constructs (e.g., FNE and FPE) in the context of interracial and same-race interactions and interpersonal evaluations presents a research gap that merits further investigation.

1.3. The present study

In our study, we aimed to examine whether FNE and FPE moderate the types of evaluations individuals make when groups vary in warmth and dominance. Hypotheses were preregistered on Open Science Framework before data collection began (see https://osf.io/ad6jq/registrations). We hypothesized that people would generally indicate a higher desire for interaction with warmer and less dominant groups (Hypothesis 1), based on prior research that people prefer to interact with warmer and less dominant people (Kelley, 1950; Rodebaugh et al., 2016).

Hypothesis 2 was that individuals with high levels of fear of evaluation (i.e., FNE and FPE) would show more desire for interaction with groups that are less warm compared to individuals with less fear of evaluation. Hypothesis 3 was that individuals with high fear of evaluation would show more desire for interaction with groups that are less dominant compared to individuals with less fear of evaluation. Hypotheses 2 and 3 were established based on previous research in social anxiety, which is positively correlated with FNE and FPE; study findings suggest that people with higher social anxiety indicated a higher desire to interact with less warm and less dominant people (i.e., people who they believed were more similar to them, since they perceive themselves to be colder and more submissive) compared to people with lower social anxiety (Rodebaugh et al., 2016).

We also aimed to determine whether examining whether racial similarity (i.e., whether the race of the person with SAD is the same as the race of the person they are interacting with) accounts for some of the effects of FNE and FPE, and whether ethnic identity plays a role in this relationship. Hypothesis 4 was that people with stronger ethnic identity would show more desire for interaction with audiences that are more similar to their race than individuals with less ethnic identity. This hypothesis was formed due to research indicating that people showed greater desire to interact with people who they believed were more similar to themselves (Rodebaugh et al., 2016). Finally, Hypothesis 5 was that fear of evaluation explains desire for interaction above and beyond racial similarity.

2. Methods

2.1. Participants

We recruited 125 Asian and 195 white participants through the undergraduate research pool at Washington University in St. Louis. They received course credit for their participation. An additional 112 participants were part of other racial groups and thus were not in the present analyses. Per our preregistration, we focused on data from our two largest racial groups, Asian and white, for this current study because we were examining racial similarity. Our sample included 235 women, 83 men, and 2 nonbinary participants. The average age of the participants was 20 years (*SD* = 1.31).

2.2. Measures

2.2.1. Brief fear of negative evaluation scale (BFNE; M. R. Leary, 1983)

The BFNE is a 12-item questionnaire that measures self-reported fear of negative evaluation. Responses are rated on a 5-point Likert-type scale from 1 (*Not at all characteristic of me*) to 5 (*Extremely characteristic of me*). The BFNE includes 8 straightforwardly-scored items and 4 reverse-scored items. For this study, only the straightforwardly-worded items were used in the total score due to evidence that they demonstrate stronger convergent validity compared to when the total score included the reverse-worded items (Rodebaugh et al., 2004; Weeks et al., 2005). Previous research (Weeks et al., 2005) also indicates that the total score of the straightforward items of the BFNE has excellent internal consistency in both control and patient samples (αs = .90 and .92). Internal consistency for the present study was excellent (α = .95).

2.2.2. Fear of positive evaluation scale (FPES; Weeks et al., 2008)

The FPES assesses worry associated with being judged favorably by others. This 10-item measure is rated on a Likert scale from 0 (*Not at all true*) to 9 (*Very true*). The two reverse-worded items in the measure were excluded from analyses, per the scale's instructions. The straightforwardly-worded items are summed to obtain a total score. FPES has been found to have good internal consistency in clinical and undergraduate samples (αs >.80), and good convergent, discriminant, and construct validity (Fergus et al., 2009; Weeks et al., 2012). We found that internal consistency was acceptable in our sample (α = .79).

2.2.3. Social interaction anxiety scale (SIAS) & social phobia scale (SPS; Mattick & Clarke, 1998)

The SIAS measures anxiety in social situations (e.g., having a conversation). The SPS measures the fear of being observed while completing routine activities (e.g., eating in front of others). These two companion scales each consist of 20 items scored from 0 (*Not at all characteristic or true of me*) to 4 (*Extremely characteristic or true of me*). The SIAS includes three reverse-scored items; these items were not included in analyses due to research that suggests that the removal of reverse-worded items improves psychometric performance (Rodebaugh et al., 2007). The SIAS straightforward items has strong construct validity and convergent validity, and high internal consistency (Rodebaugh et al., 2007, 2011), and our sample showed excellent internal consistency with the SIAS straightforward items (α = .93). The SPS has high internal consistency (α = .89) and good test-retest reliability, and we found that internal consistency was acceptable (α = .77). For this study, the total scores from the SIAS and the SPS were standardized, then combined to create a composite measure of social anxiety.

2.2.4. Beck depression inventory-II (BDI-II; Beck et al., 1996)

The BDI-II is a 21-item questionnaire that measures symptoms related to depression. The suicide item of the BDI-II was removed before data collection due to concerns that participants' suicidality would not be monitored in real-time, as this was an online study. This measure has demonstrated good concurrent validity, and excellent internal consistency (α = .92) (Beck et al., 1996). Internal consistency in our sample was excellent (α = .91).

2.2.5. International positive and negative affect schedule short form (I-PANAS-SF; Thompson, 2007)

The I-PANAS-SF is a 10-item scale intended to measure positive and negative affect. Respondents are prompted to rate the extent to which they generally feel different positive and negative affective states. Items are rated on a scale ranging from 1 (*Never*) to 5 (*Always*). The I-PANAS-SF is shown to have adequate reliability; the positive affect subscale had Cronbach's alpha of .78 and the negative affect subscale had Cronbach's alpha of .76. In our study, the positive and negative subscales both showed acceptable internal consistency (αs =.78 and .72). Responses for the positive affect items were added to create a positive affect score, and responses for the negative affect items were summed to create a negative affect score.

2.2.6. Desire for future interaction (DFI; Coyne, 1976)

The DFI is a scale that measures the willingness of the rater to engage in future social interactions with someone, such as spending more time with them, working with them, or asking them for advice. The measure consists of eight items that are rated on 5-point Likert scales ranging from 1 (*Not at all*) to 5 (*Very much*). For this study, the wording was changed to ask about the group described in the study vignettes, instead of an individual protagonist (e.g., "Would you like to meet the members of this audience again?" instead of "Would you like to meet this person again?"). The responses to the items were summed to create a total DFI score. This scale is shown to have excellent interrater reliability and internal consistency in previous studies, with an ICC of .94 and a Cronbach's alpha of .94 (Voncken et al., 2008). For our study, reliability was excellent at both the within-person ($\omega = .96$) and between-person level ($\omega = .90$).

2.2.7. Multigroup ethnic identity measure-revised (MEIM-R; Phinney & Ong, 2007)

The MEIM-R is intended to measure ethnic identity, or one's sense of self as a member of a social group. For the purposes of this measure, the authors of the scale made a distinction between racial identity and ethnic identity; they conceptualized *racial identity* as responses related to experiences of racism, whereas *ethnic identity*, as described in this measure, is about one's sense of belonging to a particular social group, which could be defined by culture, values, or traditions. We do not comment on these definitions of ethnic identity or racial identity; we intended merely to use the MEIM-R to capture *ethnic identity* as conceptualized by the measure's authors.

The measure has an open-ended question and two subscales. The open-ended question prompts the participant to self-identify as a member of a particular ethnic or racial group; the authors specified that for the purposes of this questionnaire, "it does not matter whether the label is an ethnic group or a racial group" (Phinney & Ong, 2007). The first 3-item subscale measures commitment, which refers to an individual's sense of belonging to their group; an example item is "I have a strong sense of belonging to my own ethnic group." The second 3-item subscale captures exploration, which involves an individual's efforts to seek out information and experiences related to their group; an example item is "I have spent time trying to find out more about my ethnic group, such as its history, traditions, and customs." Responses to the items are rated on a 5-point Likert-type scale from 1 (*Strongly disagree*) to 5 (*Strongly agree*). Previous research has found the MEIM-R to have good reliability ($\alpha = .88$) (Herrington et al., 2016). Phinney and Ong (2007) recommends using a total score of ethnic identity by summing all six items, and also using subscale scores if the researcher chooses to focus on just exploration or commitment of ethnic identity. For the current study, we used the total score. Internal consistency was excellent in our sample ($\alpha = .90$).

2.3. Materials and stimuli

2.3.1. Photosets

We created a series of images that each contained 12 faces of various races. The photos of faces were pulled from two publicly available databases (Strohminger et al., 2016; Mattek, Chavez, Berkowitz, Gobbini, & Chavez, 2021). The ratio of the two predominant racial groups in the sample (i.e., Asian and white) was manipulated in the images to test whether the racial makeup of groups affected participants' evaluations of others. To make the manipulation of Asian and white racial makeup less apparent, two of the 12 people were always individuals of races other than Asian and white (e.g., Black). The ratio of the photos varied from majority white to majority Asian.

We developed two different photosets, Set A and Set B, to systematically examine whether the specific combination of faces in a given image affected participants' evaluations of others. The ratio of the predominant racial groups in Photoset A and Photoset B remained the same, but the images of the 12 individual faces of different races were featured in different combinations. There were seven photos in each photoset (see https://osf.io/ad6jq/ to view stimuli).

2.3.2. Vignettes

We created eight vignettes describing groups of people based on the different octants of the interpersonal circumplex; in other words, the descriptions in the vignettes systematically varied in warmth and dominance (see https://osf.io/ad6jq/ for the vignettes).

2.3.2.1. Manipulation check.

To examine whether the vignettes varied in warmth and dominance as intended, we ran a pilot study in which participants rated the vignettes on warmth and dominance. The participants ($N = 286$) were also recruited through the undergraduate research pool at Washington University in St. Louis; however, none of the participants for the pilot study overlapped with participants in the current study. Similar to the methods described in a previous study (Rodebaugh et al., 2016), participants were asked to rate the perceived warmth (on a 5-point scale ranging from *cold* to *warm*) and perceived dominance (on a 5-point scale from *submissive* to *dominant*) of all 8 vignettes, which were presented in a randomized order. We used a multilevel model in which participants' ratings of perceived warmth and perceived dominance were predicted by the intended warmth and intended dominance of each vignette. We plotted the estimates from the model onto a circumplex to compare the average position of the participants' perceived warmth and dominance ratings to the intended position of the vignettes (see Fig. 1).

As conveyed in the figure, participants showed a general pattern similar to the circumplex; however, their ratings did not match the circumplex exactly. Participants rated the vignettes as warmer than intended for the colder vignettes. Participants also rated the dominant octants as less dominant than intended, and the submissive octants as more dominant than intended. However, all of the vignettes remained in the general category as the intended octants (e.g., the cold dominant vignette was in the appropriate octant and did not switch to the warm submissive or the cold submissive octants).

2.4. Procedure

After reviewing an IRB-approved information sheet, participants completed a series of questionnaires via Qualtrics, including some measures that are not described for this present project. Of relevance to this study, the battery included a demographics questionnaire, the BFNE, FPES, SIAS, SPS, BDI, and I-PANAS-SF.

After completing the questionnaires, participants completed the photo-vignette task on Qualtrics. Participants were randomly assigned to see images from either Photoset A or Photoset B. During the photo-vignette task, participants read vignettes that were randomly paired with images that each included 12 faces of various races. Each participant was presented with a randomized seven out of eight possible vignettes; paired with each vignette was a random image from the photoset group to which they were randomly assigned. The vignettes included descriptions of the group that the individual was interacting with; the descriptions of the group's behavior varied in warmth and dominance. For example, part of the *warm* group's vignette illustrates that "the audience members are making eye contact with you and smiling." On the other hand, the *cold submissive* group members "avoid eye contact and quickly look away" and "don't respond to you when you thank them."

Participants saw seven photo-vignette pairings and rated the desire for future interaction (DFI) with that group described in the photo-vignette pairing. After participants finished the photo-vignette task, they completed the MEIM-R. Questions about ethnic identity were asked after the photo-vignette task instead of before to reduce the likelihood of participants noticing that racial similarity was being manipulated in the photo-vignettes that they were presented.

2.5. Statistical analysis

We used multilevel modeling (MLM) with random effects and data nested by participant. The variables at the within level (Level 1) included the intended warmth and intended dominance associated with each vignette that the participant rated; these intended warmth and dominance ratings of each octant were the same assignments used in the pilot study and were identical across participants. The variables at the between level (Level 2) included social anxiety, FNE, FPE, depression, positive affect, negative affect, racial similarity, and ethnic identity. All the Level 1 and Level 2 variables were standardized (i.e., with a mean of 0 and a standard deviation of 1) prior to entering them in the models. We used the lme4 package in R to fit our models.

3. Results

3.1. Descriptives for DFI ratings

Overall DFI ratings are in Table 1. It was notable that DFI ratings showed more variance across warmer octants.

3.2. Hypothesis 1

Consistent with our hypothesis, we found that participants showed a greater desire for future interaction with warmer groups ($\beta = 4.79$, $p < .001$, 95% CI [4.45, 5.13]). Furthermore, participants showed lower DFI with groups that were more dominant ($\beta = -1.02$, $p < .001$, 95% CI [-1.27, $-.78$]). The interaction effects of warmth and dominance were also significant ($\beta = -.44$, $p < .001$, 95% CI [$-.67$, $-.21$]); in other words, dominance affected the way warmth predicted DFI, in that dominance attenuated the effects of warmth.

3.3. Hypotheses 2 and 3

Contrary to Hypothesis 2, we found that at the average level of FNE, people demonstrated higher DFI for groups that were warmer, and this relationship grew stronger as FNE increased, as shown in Table 2 and Fig. 2. For Hypothesis 3, we found that at the average level of FPE, people reported lower DFI for more dominant groups, and the slope became steeper as FPE increased, as shown in Table 2 and Fig. 3, which was generally consistent with our hypothesis.

Table 1
Descriptive Data of Participants' Desire for Interaction Ratings.

Octant	M (SD)
Cold	4.49 (5.44)
Cold Dominant	5.09 (5.52)
Cold Submissive	6.77 (5.91)
Dominant	9.44 (6.50)
Submissive	11.04 (6.90)
Warm Dominant	13.67 (8.26)
Warm Submissive	17.63 (8.13)
Warm	18.62 (7.74)

Table 2
Results from Multilevel Model for Hypothesis 2 and 3.

	Desire for Future Interaction		
Predictors	*β*	*CI*	*p*
(Intercept)	10.89	10.39 – 11.39	**<.001**
Warmth	4.78	4.44 – 5.12	**<.001**
Dominance	-1.01	-1.25 – -0.77	**<.001**
FNE	0.41	-0.16 – 0.99	.159
FPE	-0.13	-0.70 – 0.45	.665
Warmth*Dominance	-0.43	-0.66 – -0.20	**<.001**
Warmth*FNE	0.46	0.07 – 0.85	**.020**
Warmth*FPE	0.03	-0.35 – 0.42	.866
Dominance*FNE	0.15	-0.13 – 0.43	.286
Dominance*FPE	-0.39	-0.66 – -0.11	**.006**
Warmth*Dominance*FNE	0.18	-0.09 – 0.45	.184
Warmth*Dominance*FPE	-0.25	-0.52 – 0.02	.069

3.4. Follow-up analyses

As previously mentioned, FNE and FPE are thought to be core constructs of SAD (Weeks et al., 2010). Furthermore, FPE, FNE, and SAD all relate positively and uniquely to trait negative affect; there is also some evidence to suggest that FPE is uniquely related to trait positive affect (Weeks & Howell, 2012). Thus, following our pre-registered analysis plan, we conducted follow-up analyses to see whether the effects we observed in our models were due to social anxiety or positive or negative affect rather than FNE and FPE. We also tested whether depression accounted for these effects because of the high comorbidity between SAD and depression (Kessler et al., 2005).

To test whether the effects we observed were explained by social anxiety, depression, negative affect or positive affect, we included each of these variables in separate models with intended warmth, intended dominance, FNE, and FPE. Even after including social anxiety, depression, negative affect, and positive affect in separate models, all of the effects we observed remained statistically significant, suggesting that these variables do not explain our findings better than FNE and FPE.

3.5. Hypothesis 4

To explore the relationship between ethnic identity and racial similarity to DFI (Hypothesis 4), we examined a multilevel model in which racial similarity (i.e., the percentage of individuals in a photo that are of the same race as the respondent) was used to predict DFI at Level 1. The Level 2 variable of ethnic identity was added to the model, as was the cross-level interaction between ethnic identity and racial similarity, to predict DFI.

We did not find an effect of racial similarity, ethnic identity, or the interaction between racial similarity and ethnic identity on DFI (*ps* $> .05$). In other words, we did not find support for our hypothesis that people with stronger ethnic identity showed more DFI with audiences that are more similar to their race than individuals with less ethnic identity.

3.6. Follow-up analyses

In our exploratory follow-up analyses, we tested whether participants' race affected our findings for Hypothesis 4. We added race to Level 2 of the multilevel model and examined the interactions between race and our variables of interest. Analyses showed that the interaction between ethnic identity and race was significant ($\beta = 1.18$, $p = .044$, 95% CI [.03, 2.34]), such that the effect of ethnic identity (i.e., one's sense of belonging to a particular social group based on culture, values, or traditions) on DFI is stronger in white participants than in Asian participants. However, we did not attempt to interpret our findings because it is difficult to conceptualize what aspects of white ethnic identity, such as traditions or culture, mean in practice.

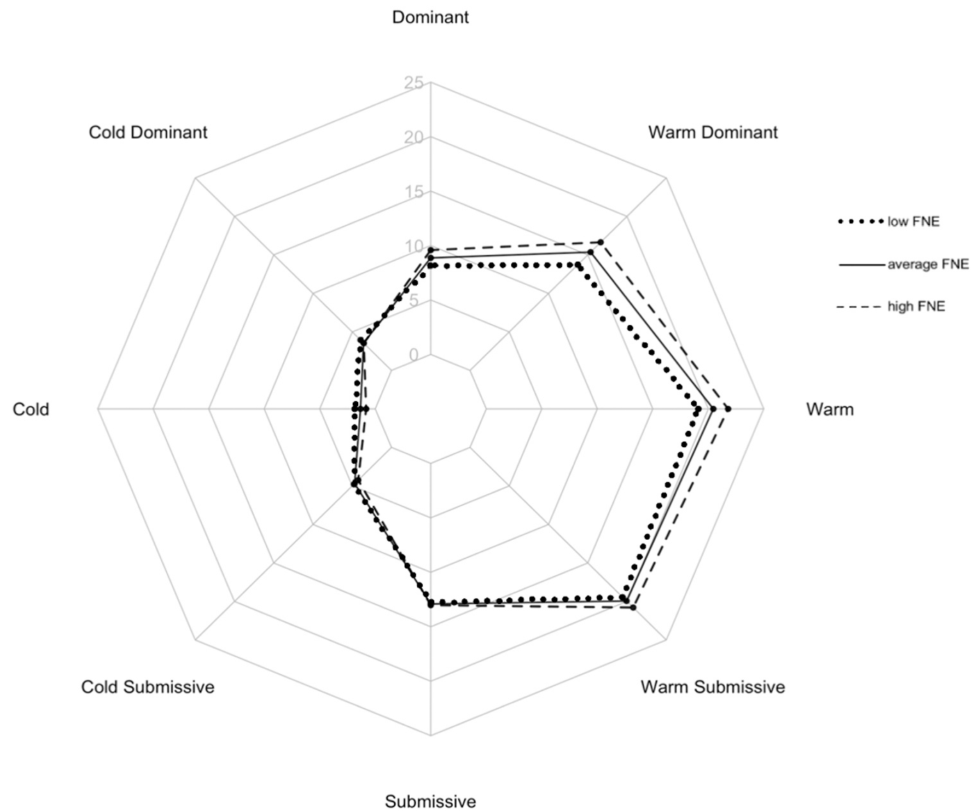

Fig. 2. *Desire for Future Interaction for Individuals with Average FPE and Varying FNE*. Note. This plot shows desire for interaction based on level of FNE. The labels on the axes ranging from 0 to 25 represent the desire for interaction score. The points on the graph that are closer to the outer edge of the plot, toward the octant label, reflect greater desire for interaction. The values that are shown here ranging from 0 to 25 reflect the estimated values, not the full range of the DFI scale; the full scale ranges from 0 to 32.

3.7. *Hypothesis 5*

To address our final hypothesis, we used MLM to examine whether fear of evaluation explains DFI above and beyond ethnic similarity. We first regressed DFI onto racial similarity at Level 1. Our Level 2 variables (FNE, FPE) and their interactions with our Level 1 predictors (racial similarity, intended warmth, intended dominance) were then added to our model.

The relationship between FNE, FPE, and DFI remained the same even after including racial similarity. Based on our model, we also saw that racial similarity did not predict DFI, and this did not depend on FNE or FPE. Our analyses did show a significant three-way interaction between warmth, dominance, and FPE ($\beta = -0.29$, $p = .026$, 95% CI [−.55, −.04]). In other words, the relationship between warmth and DFI was moderated by dominance at higher levels of FPE and not at lower levels of FPE; at higher FPE, dominance had an effect on the relationship between warmth and DFI, such that warmth related to higher DFI much more for less dominant groups than more dominant groups. However, we interpret this result with caution because this interaction was neither predicted nor significant in the previous analyses for Hypotheses 2 and 3. See Table 3 for details.

4. Discussion

We examined whether FNE and FPE affect the way individuals perceive groups in the context of varying perceived racial similarity. Consistent with past research, we found that individuals in general prefer interactions with warmer (Kelley, 1950; Rodebaugh et al., 2016) and less dominant people (Rodebaugh et al., 2016). We also found that people with higher FPE more strongly preferred less dominant groups than people with lower FPE, consistent with the psychoevolutionary perspective that individuals with higher FPE avoid conflict with people of higher social rank (Gilbert, 2001; Trower & Gilbert, 1989). Contrary to our hypothesis, FNE did not show such an interaction with dominance. Based on psychoevolutionary theory, one might expect that someone who is higher in FNE (i.e., fears rejection from the group by being deemed “not good enough”) would have a greater preference for less dominant individuals because more submissive behaviors from others would provide reassurance that the individual is not in danger of being rejected. Our results seem to conflict with this account. In contrast, we did find an unexpected relationship between FNE, warmth, and DFI, such that individuals with higher FNE showed greater preference for warmer groups. Although this finding may be intuitive, we did not hypothesize this effect because past research suggests that individuals higher in social anxiety show greater desire to interact with colder individuals (Rodebaugh et al., 2016).

A possible explanation for this relationship of FNE and warmth on DFI is that affiliative cues can encourage a sense of safety as a member within a group. Trower and Gilbert (1989) theorized that in addition to the *social rank system*, which is sensitive to information regarding social hierarchy, humans developed an *affiliation system* that is sensitive to cues indicating cooperation and support; thus, if group members send affiliative signals, individuals may be more likely to exhibit approach behavior because they feel safe (Trower & Gilbert, 1989). Given our findings, it is possible that warmer groups provide more reassurance to individuals, especially to those with higher FNE, and thus those individuals are more likely to have future interactions with them.

However, although Trower and Gilbert (1989) proposed that humans in general have developed an affiliation system, they also suggested that individuals with SAD are less sensitive to affiliation cues and that they are overly concerned about cues regarding social rank (Gilbert, 2001). If this were the case, one might expect that individuals high in FNE would

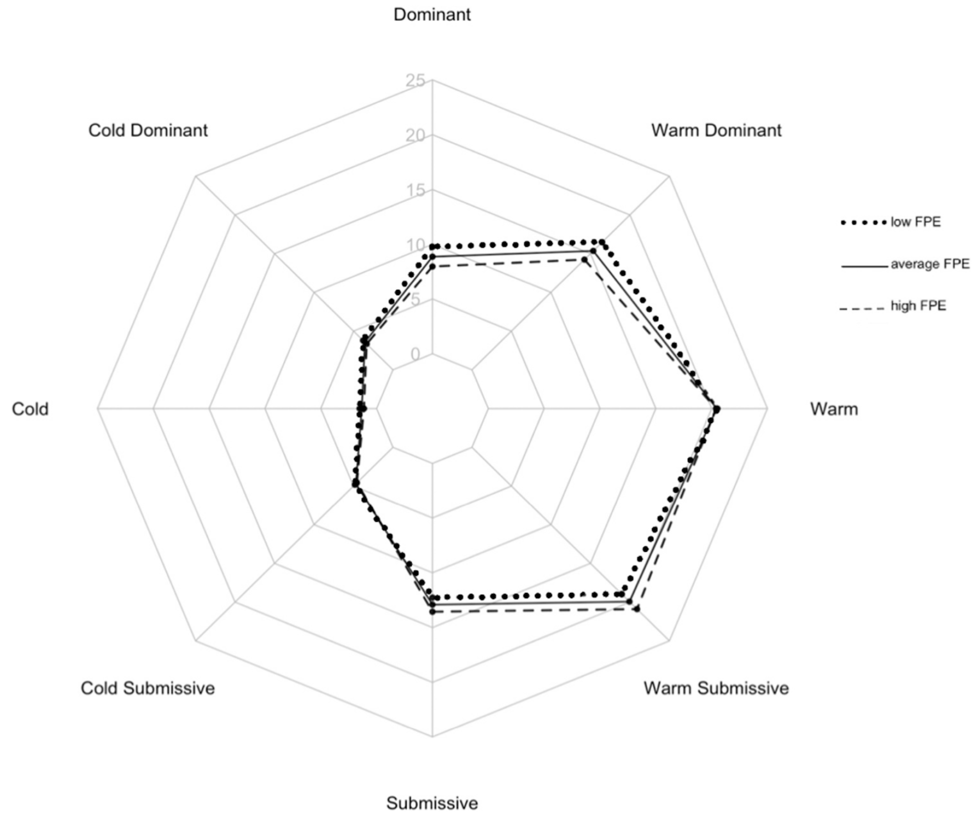

Fig. 3. *Desire for Future Interaction for Individuals with Average FNE and Varying FPE.* Note. This plot shows desire for interaction based on level of FPE. The labels on the axes ranging from 0 to 25 represent the desire for interaction score. The points on the graph that are closer to the outer edge of the plot, toward the octant label, reflect greater desire for interaction. The values that are shown here ranging from 0 to 25 reflect the estimated values, not the full range of the DFI scale; the full scale ranges from 0 to 32.

Table 3
Results from Multilevel Model for Hypothesis 5.

	Desire for Future Interaction		
Predictors	*β*	*CI*	*p*
(Intercept)	10.87	10.37 – 11.37	**<.001**
Racial Similarity	-0.01	-0.23 – 0.22	.965
Warmth	4.78	4.44 – 5.12	**<.001**
Dominance	-1.00	-1.24 – -0.76	**<.001**
FNE	0.43	-0.14 – 1.01	.138
FPE	-0.12	-0.70 – 0.45	.675
Racial Similarity*Warmth	0.01	-0.22 – 0.24	.938
Racial Similarity*Dominance	-0.14	-0.37 – 0.09	.234
Warmth*Dominance	-0.42	-0.64 – -0.19	**<.001**
Racial Similarity*FNE	0.19	-0.07 – 0.45	.152
Warmth*FNE	0.46	0.07 – 0.84	**.021**
Dominance*FNE	0.15	-0.13 – 0.43	.297
Racial Similarity*FPE	-0.05	-0.31 – 0.21	.722
Warmth*FPE	0.02	-0.37 – 0.41	.917
Dominance*FPE	-0.39	-0.67 – -0.11	**.006**
Racial Similarity*Warmth*Dominance	0.12	-0.15 – 0.38	.389
Racial Similarity*Warmth*FNE	0.15	-0.11 – 0.41	.245
Racial Similarity*Dominance*FNE	0.19	-0.08 – 0.46	.167
Warmth*Dominance*FNE	0.18	-0.08 – 0.44	.178
Racial Similarity*Warmth*FPE	-0.16	-0.43 – 0.10	.231
Racial Similarity*Dominance*FPE	0.18	-0.09 – 0.45	.198
Warmth*Dominance*FPE	-0.29	-0.55 – -0.04	**.026**
Racial Similarity*Warmth*Dominance*FNE	-0.10	-0.40 – 0.20	.504
Racial Similarity*Warmth*Dominance*FPE	0.12	-0.19 – 0.43	.445

not show a bias for warmth, yet our results suggest otherwise. Furthermore, as we mentioned previously, we did not find support for a relationship between FNE and dominance on desire for future interaction. This discrepancy between our results and Trower and Gilbert's theory (1989) may be because individuals who fear evaluation may be more sensitive to affiliation cues than originally proposed. Another possibility is that individuals with high FNE may respond differently to signals of warmth when they are engaging in group interactions instead of one-on-one interactions. This seems plausible when considering that evolutionarily, one might fear negative evaluation from a group differently than negative evaluation from an individual, because rejection from one's group may pose a more severe threat to one's safety than rejection from one person. The research thus far has focused on affiliation and DFI within interactions between individuals (Aderka et al., 2013; Haker et al., 2014; Rodebaugh et al., 2016), so further investigation of the relationship between SAD, affiliation, and DFI regarding groups is needed to draw firmer conclusions.

Despite the need for continued research, our findings that FNE is related to warmth and that FPE is related to dominance align with the growing body of research that FNE and FPE are separate, though related, constructs that are each important to social anxiety (Cook et al., 2022). If FNE and FPE are related to different domains of interpersonal evaluation, this would have important clinical implications when addressing interpersonal impairment. For example, a person high in FPE may benefit from treatment (e.g., exposure) that focuses on interacting with people of different levels of dominance, whereas a person who is high in FNE may benefit from treatment targeting coldness versus warmth. An individual who is high in both FNE and FPE may benefit from treatment addressing both warmth and dominance. Further investigation on the relationships between FNE, FPE, warmth, and dominance in interpersonal evaluations could help clinicians tailor treatment more effectively.

Regarding the relationship between racial similarity, ethnic identity, and DFI, we did not find support for our hypothesis that people with higher ethnic identity showed greater DFI for groups that are more similar to their race compared to individuals with lower ethnic identity.

A possible reason for this can be found in the social diversity literature; research shows that more social diversity is associated with a decreased likelihood of differentiating ethnic groups on different dimensions, including warmth. More specifically, students in more diverse campuses perceived greater similarity, as opposed to greater differentiation, among ethnic groups (Bai et al., 2020). The undergraduate participants for the present study were selected from a campus that has a large percentage of Asian (33.88%) and white (47.58%) students (Washington University in St. Louis, 2021). It is plausible that the sample in this study perceived more similarity between themselves and the groups shown in the study stimuli and were less likely to differentiate others using characteristics of race or ethnic identity, because of their increased exposure to diversity on campus. If the participants from our sample were not particularly influenced by race due to their regular exposure to Asian and white peers, it is possible that our objective to investigate the relationship between interpersonal evaluation, racial similarity, and ethnic identity was only partially addressed and should be examined further. For example, would these effects be present if the racial makeup of the groups included people from racial groups that were not as prevalent on campus, such as Black individuals? More research to examine these effects with participants from different communities varying in diversity, and also including participants of other races besides white and Asian, could further clarify the relationship between racial similarity, ethnic identity, and DFI.

4.1. Limitations

Our results should be interpreted in the context of several limitations. For one, our study used vignettes and images to simulate interactions between individuals and groups, and vignettes and photos may not reflect interpersonal behavior in real-life settings. Additionally, the type of interaction that was examined in this study was specifically exchanges between individuals and groups in the context of a class presentation. However, there are many different types of interpersonal exchanges between an individual and others (e.g., shared meals, group activities); it is possible that the type of group interactions where the individual is clearly in the center of attention (e.g., a presentation) versus a group interaction where the attention may be less focused on the individual (e.g., a holiday party) may affect the types of evaluations that an individual makes. Furthermore, social contexts in which a person's status or level of inclusion within a group changes (e.g., joining a new team of supervisors and coworkers) compared to situations in which an individual's position in the group is stable, may be particularly impactful in regard to fear of evaluation. Additionally, although the DFI measure covers a range of different interactions (e.g., "Would you like to work with these people?", "Would you consider having them for housemates?"), arguably there are more obviously status-focused situations, such as working under someone or leading a team, that are not captured by the DFI. For these reasons, it may be helpful for future work to address a wider range of interpersonal group contexts.

An important consideration is that our analyses included data from only Asian and white participants because the study manipulated the percentage of Asian and white individuals shown in our stimuli to examine the dimension of racial similarity. For this reason, our results do not necessarily apply to individuals of other racial groups. Research shows that people of different races (e.g., Asian American compared to African American) are perceived differently on multiple dimensions, including status and warmth (Fiske et al., 2002; Zou & Cheryan, 2017). Therefore, it is possible that the results from our study, which used stimuli showing Asian and white individuals, may not be the same when using stimuli showing people of other races, such as Black individuals, if the perceptions of various racial groups differ. Therefore, studies manipulating racial similarity to other racial groups, and including data from participants beyond those that identify as white or Asian, would provide further insight on the effects we observed.

Despite limitations, the findings from this study are a step toward expanding our current understanding of interpersonal evaluations in SAD, which have largely focused on one-on-one interactions until this point. Overall, our results suggest that FNE may play a role in interpersonal evaluations of warmth in group interactions, and FPE may play a role in evaluations of dominance. These findings contribute to the growing literature that FNE and FPE are separable constructs that are both important facets to social anxiety and have implications for treatment in SAD.

Declaration of Competing Interest

None.

References

Aderka, I. M., Haker, A., Marom, S., Hermesh, H., & Gilboa-Schechtman, E. (2013). Information-seeking bias in social anxiety disorder. *Journal of Abnormal Psychology, 122*(1), 7.

American Psychiatric Association. (2022). *Diagnostic and statistical manual of mental disorders (5th ed., text rev.)*. https://doi.org/10.1176/appi.books.9780890425787.

Bai, X., Ramos, M. R., & Fiske, S. T. (2020). As diversity increases, people paradoxically perceive social groups as more similar. *Proceedings of the National Academy of Sciences, 117*(23), 12741–12749.

Beck, A. T., Steer, R. A., & Brown, G. K. (1996). *Manual for the Beck Depression Inventory-II*. San Antonio, TX: Psychological Corporation.

Brewer, M. B. (2017). Intergroup Discrimination: Ingroup Love Or Outgroup Hate?. *The Cambridge handbook of the psychology of prejudice* (pp. 90–110). Cambridge University Press. https://doi.org/10.1017/9781316161579.005

Cook, S. I., Moore, S., Bryant, C., & Phillips, L. J. (2022). The role of fear of positive evaluation in social anxiety: A systematic review and meta-analysis. *Clinical Psychology: Science and Practice, 29*(4), 352–369. https://doi.org/10.1037/cps0000082

Coyne, J. C. (1976). Depression and the response of others. *Journal of Abnormal Psychology, 85*(2), 186.

Fergus, T. A., Valentiner, D. P., McGrath, P. B., Stephenson, K., Gier, S., & Jencius, S. (2009). The fear of positive evaluation scale: Psychometric properties in a clinical sample. *Journal of Anxiety Disorders, 23*(8), 1177–1183. https://doi.org/10.1016/j.janxdis.2009.07.024

Fiske, S. T., Cuddy, A. J., Glick, P., & Xu, J. (2002). A model of (often mixed) stereotype content: Competence and warmth respectively follow from perceived status and competition. *Journal of Personality and Social Psychology, 82*(6), 878.

Gilbert, P. (2001). Evolution and social anxiety: The role of attraction, social competition, and social hierarchies. *Psychiatric Clinics, 24*(4), 723–751.

Haker, A., Aderka, I. M., Marom, S., Hermesh, H., & Gilboa-Schechtman, E. (2014). Impression formation and revision in social anxiety disorder. *Journal of Anxiety Disorders, 28*(2), 133–139. https://doi.org/10.1016/j.janxdis.2013.05.001

Hamley, L., Houkamau, C. A., Osborne, D., Barlow, F. K., & Sibley, C. G. (2020). Ingroup love or outgroup hate (or both)? Mapping distinct bias profiles in the population. *Personality and Social Psychology Bulletin, 46*(2), 171–188. https://doi.org/10.1177/0146167219845919

Herrington, H. M., Smith, T. B., Feinauer, E., & Griner, D. (2016). Reliability generalization of the Multigroup Ethnic Identity Measure-Revised (MEIM-R). *Journal of Counseling Psychology, 63*(5), 586–593. https://doi.org/10.1037/cou0000148

Jager, P. D., Suliman, S., & Seedat, S. (2014). Role of ethnicity in social anxiety disorder: A cross-sectional survey among health science students. *World Journal of Clinical Cases: WJCC, 2*(7), 265–271. https://doi.org/10.12998/wjcc.v2.i7.265

Kelley, H. (1950). *The warm-cold variable in first impressions of persons*.

Kessler, R. C., Berglund, P., Demler, O., Jin, R., Merikangas, K. R., & Walters, E. E. (2005). Lifetime prevalence and age-of-onset distributions of DSM-IV disorders in the National Comorbidity Survey Replication. *Archives of General Psychiatry, 62*, 593–602. https://doi.org/10.1001/archpsyc.62.6.593

Leary, M. R. (1983). A brief version of the Fear of Negative Evaluation Scale. *Personality and Social Psychology Bulletin, 9*(3), 371–375. https://doi.org/10.1177/0146167283093007

Leary, T. (1957). *Interpersonal diagnosis of personality*. Ronald Press.

Mattick, R. P., & Clarke, J. C. (1998). Development and validation of measures of social phobia scrutiny fear and social interaction anxiety. *Behaviour Research and Therapy, 36*(4), 455–470. https://doi.org/10.1016/S0005-7967(97)10031-6

Mattek, A., Chavez, S., Berkowitz, J. L., Gobbini, M. I., & Chavez, R. (2021, October 2). 100 Peers: An open set of controlled emotional facial expressions for use in psychology and neuroscience experiments. Retrieved from psyarxiv.com/njvts.

Okawa, S., Arai, H., Sasagawa, S., Ishikawa, S., Norberg, M. M., Schmidt, N. B., Kwon, J.-H., Rapee, R. M., & Shimizu, E. (2021). A cross-cultural comparison of the bivalent fear of evaluation model for social anxiety. *Journal of Behavioral and Cognitive Therapy, 31*(3), 205–213. https://doi.org/10.1016/j.jbct.2021.01.003

Phinney, J. S., & Ong, A. D. (2007). Conceptualization and measurement of ethnic identity: Current status and future directions. *Journal of Counseling Psychology, 54*(3), 271.

Rodebaugh, T. L., Bielak, T., Vidovic, V., & Moscovitch, D. A. (2016). The effects of social anxiety on interpersonal evaluations of warmth and dominance. *Journal of Anxiety Disorders, 38*, 68–78.

Rodebaugh, T. L., Heimberg, R. G., Brown, P. J., Fernandez, K. C., Blanco, C., Schneier, F. R., & Liebowitz, M. R. (2011). More reasons to be straightforward: Findings and norms for two scales relevant to social anxiety. *Journal of Anxiety Disorders, 25*(5), 623–630. https://doi.org/10.1016/j.janxdis.2011.02.002

Rodebaugh, T. L., Weeks, J. W., Gordon, E. A., Langer, J. K., & Heimberg, R. G. (2012). The longitudinal relationship between fear of positive evaluation and fear of negative evaluation. *Anxiety, Stress & Coping, 25*(2), 167–182. https://doi.org/10.1080/10615806.2011.569709

Rodebaugh, T. L., Woods, C. M., & Heimberg, R. G. (2007). The reverse of social anxiety is not always the opposite: The reverse-scored items of the Social Interaction Anxiety Scale do not belong. *Behavior Therapy, 38*(2), 192–206. https://doi.org/10.1016/j.beth.2006.08.001

Rodebaugh, T. L., Woods, C. M., Thissen, D. M., Heimberg, R. G., Chambless, D. L., & Rapee, R. M. (2004). More information from fewer questions: The factor structure and item properties of the original and brief fear of negative evaluation scale. *Psychological Assessment, 16*(2), 169.

Ruscio, A. M., Brown, T. A., Chiu, W. T., Sareen, J., & Kessler, R. C. (2008). Social fears and social phobia in the United States: Results from the National Comorbidity Survey Replication. *Psychological Medicine, 38*, 15–28. https://doi.org/10.1017/S0033291707001699

Sidanius, J., & Pratto, F. (1999). *Social dominance: An intergroup theory of social hierarchy and oppression*. Cambridge University Press.

Strohminger, N., Gray, K., Chituc, V., Heffner, J., Schein, C., & Heagins, T. B. (2016). The MR2: A multi-racial, mega-resolution database of facial stimuli. *Behavior Research Methods, 48*, 1197–1204.

Thompson, E. R. (2007). Development and validation of an internationally reliable short-form of the Positive and Negative Affect Schedule (PANAS). *Journal of Cross-Cultural Psychology, 38*(2), 227–242. https://doi.org/10.1177/0022022106297301

Tonge, N. A., Lim, M. H., Piccirillo, M. L., Fernandez, K. C., Langer, J. K., & Rodebaugh, T. L. (2020). Interpersonal problems in social anxiety disorder across different relational contexts. *Journal of Anxiety Disorders, 75*, Article 102275. https://doi.org/10.1016/j.janxdis.2020.102275

Trower, P., & Gilbert, P. (1989). New theoretical conceptions of social anxiety and social phobia. *Clinical Psychology Review, 9*(1), 19–35.

Voncken, M. J., Alden, L. E., Bögels, S. M., & Roelofs, J. (2008). Social rejection in social anxiety disorder: The role of performance deficits, evoked negative emotions and dissimilarity. *British Journal of Clinical Psychology, 47*(4), 439–450.

Wang, W.-T., Hsu, W.-Y., Chiu, Y.-C., & Liang, C.-W. (2012). The hierarchical model of social interaction anxiety and depression: The critical roles of fears of evaluation. *Journal of Anxiety Disorders, 26*(1), 215–224.

Washington University in St. Louis. (2021). *Student Body Diversity*. Office of the University Registrar. 〈https://registrar.wustl.edu/student-information-systems/student-enrollment-and-graduation-statistics/student-body-diversity/〉.

Weeks, J. W., Heimberg, R. G., Fresco, D. M., Hart, T. A., Turk, C. L., Schneier, F. R., & Liebowitz, M. R. (2005). Empirical validation and psychometric evaluation of the Brief Fear of Negative Evaluation Scale in patients with social anxiety disorder. *Psychological Assessment, 17*(2), 179.

Weeks, J. W., Heimberg, R. G., & Rodebaugh, T. L. (2008). The Fear of Positive Evaluation Scale: Assessing a proposed cognitive component of social anxiety. *Journal of Anxiety Disorders, 22*(1), 44–55. https://doi.org/10.1016/j.janxdis.2007.08.002

Weeks, J. W., Heimberg, R. G., Rodebaugh, T. L., Goldin, P. R., & Gross, J. J. (2012). Psychometric evaluation of the fear of positive evaluation scale in patients with social anxiety disorder. *Psychological Assessment, 24*(2), 301. https://doi.org/10.1037/a0025723

Weeks, J. W., & Howell, A. N. (2012). The bivalent fear of evaluation model of social anxiety: Further integrating findings on fears of positive and negative evaluation. *Cognitive Behaviour Therapy, 41*(2), 83–95. https://doi.org/10.1080/16506073.2012.661452

Weeks, J. W., Jakatdar, T. A., & Heimberg, R. G. (2010). Comparing and contrasting fears of positive and negative evaluation as facets of social anxiety. *Journal of Social and Clinical Psychology, 29*(1), 68–94. https://doi.org/10.1521/jscp.2010.29.1.68

Weisman, O., Aderka, I. M., Marom, S., Hermesh, H., & Gilboa-Schechtman, E. (2011). Social rank and affiliation in social anxiety disorder. *Behaviour Research and Therapy, 49*(6), 399–405. https://doi.org/10.1016/j.brat.2011.03.010

Wiggins, J. S. (1979). A psychological taxonomy of trait-descriptive terms: The interpersonal domain. *Journal of Personality and Social Psychology, 37*(3), 395.

Zou, L. X., & Cheryan, S. (2017). Two axes of subordination: A new model of racial position. *Journal of Personality and Social Psychology, 112*(5), 696.